Year 13 Maths - Further differentiation

Similar questions - Implicit and parametric differentiation

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A curve has equation $x^2 + 3xy + y^2 = 5$.

(a) Show that $\dfrac{\mathrm{d}y}{\mathrm{d}x} = -\dfrac{2x+3y}{3x+2y}$. **[3]**

(b) Find the equation of the tangent to the curve at the point $(1,1)$. **[3]**

(c) Show that there is no point on the curve at which the tangent is parallel to the y-axis. **[4]**

2 A curve has parametric equations $x = t^2 + 1$, $y = t^3 - 3t$.

(a) Find $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ in terms of t. **[3]**

(b) Find the coordinates of the stationary points of the curve. **[3]**

(c) Show that the Cartesian equation of the curve is $y^2 = (x-1)(x-4)^2$. **[4]**

(d) Find the gradient of the curve at the point where $t = 2$. **[1]**

Question 1

(a) Differentiating implicitly: $2x+3y+3x\dfrac{\mathrm{d}y}{\mathrm{d}x}+2y\dfrac{\mathrm{d}y}{\mathrm{d}x}=0$, so $\dfrac{\mathrm{d}y}{\mathrm{d}x}=-\dfrac{2x+3y}{3x+2y}$

(b) At $(1,1)$: $-\dfrac{5}{5}=-1$, so $y-1=-(x-1)$, i.e. $y=2-x$

(c) Parallel to the y-axis means $3x+2y=0$, so $y=-\frac{3}{2}x$. Substituting: $x^2-\frac{9}{2}x^2+\frac{9}{4}x^2=5$, so $-\frac{5}{4}x^2=5$ and $x^2=-4$, which has no real solution

Question 2

(a) $\dfrac{\mathrm{d}y}{\mathrm{d}t}=3t^2-3$, $\dfrac{\mathrm{d}x}{\mathrm{d}t}=2t$. $\dfrac{\mathrm{d}y}{\mathrm{d}x}=\dfrac{3(t^2-1)}{2t}$

(b) Zero when $t=\pm1$: $(2,-2)$ and $(2,2)$

(c) $t^2=x-1$. $y^2=t^2(t^2-3)^2=(x-1)(x-1-3)^2=(x-1)(x-4)^2$

(d) $\dfrac{3(4-1)}{4}=\dfrac{9}{4}$